

SHRIKANT G PILLAMARAPU

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Education

Master of Science in Electrical Engineering

Texas Tech University, Lubbock, Texas

GPA: 3.58

Graduation Date: May 2017

Bachelor of Engineering in Electronics and Telecommunication

Mumbai University, India

GPA: 3.4

Graduation Date: May 2014

Certifications & Scholarships

- Teradyne Semiconductor Test Training certification, Nov 2017
- NI Certified LabVIEW Associate Developer, Serial Number: 100-316-14388
- Recipient of TI Semiconductor Prod Engineer Scholarship, May 2016-May 2017
- Recipient of TTU Seacat EE Scholarship, July 2015-May 2017

Experience

Senior Test Development Engineer

Mar 2020 - Mar 2025

Analog Devices, Durham, North Carolina

- Led the team in delivering the initial silicon bring-up and designing production test programs for several NPI products on National Instruments STS and Teradyne Ultraflex automated test equipment.
- Designed and implemented TSSI Wavemaker Plus scenarios for translating WGL patterns to ATE platforms and converting VCD to ATE-compatible formats.
- Converted Scan, Memory BIST, and JTAG test patterns using the TSSI Wavemaker Plus tool.
- Developed and tested test routines for digital blocks in mixed-signal ICs using NI tools.
- Executed SPI protocol characterization on ATE, identifying critical timing parameters like setup and hold times and mitigating timing violations, and created Python scripts for SPI read and write operations.
- Analyzed STDF datalogs extensively to drive yield improvements, identified failure signatures in digital pattern failures, and optimized fallout debugging methods.
- Achieved faster digital scan block performance by conducting validations across temperature ranges and on nominal, slow, and fast skew units, reducing overall test time.
- Contributed to pre-silicon digital pin layout design, focusing on scan-based pins to enhance scan compression, tester memory efficiency, and overall test time.
- Collaborated with cross-functional teams, including DFT, DV, and Design, to address critical issues during the NPI production ramp.
- Proficient in utilizing Perforce to manage and version control ATE patterns and scenarios used by the digital test team.
- Trained and mentored interns and newly graduated team members, facilitating their integration into multiple project workflows.

Test Development Engineer

Jul 2017 - Jan 2020

ON Semiconductor, Boise, Idaho

- Directed all aspects of testing as Test Lead, with a focus on test deliverables, first-silicon validation, and bring-up processes.
- Created and implemented test code and text-based programs with VBA and Python, leveraging the IG-XL platform on IP750EX testers.
- Organized and led design review discussions for load boards and probe cards, aiming for increased parallelism.
- Developed expertise in DFT techniques, MBIST, and Scan patterns, while collaborating with Design, DFT, Process, and Product Engineers to resolve issues documented in JIRA.
- Led the bring-up of test programs across varying temperature flows, enabling seamless progression from sampling to mass production.
- Performed data correlation for new product tests on DataPower and Optimal Plus, guided by Six Sigma practices.
- Conducted training sessions for interns and new hires on company-specific tools, software, and standardized testing protocols.
- Handled RMA issues end-to-end, from root cause analysis to closure, supported by detailed 8D reports.
- Contributed to team code collaboration and ensured smooth integration into active projects using GitLab and SVN Repository.

Lab Assistant

May 2016 - May 2017

Texas Tech University, Lubbock, Texas

Program for Semiconductor Product Engineering (PSPE) Lab

- Designed and implemented test stand sequences in LabVIEW to test Digital ICs and data converters on NI STS T4, PXIe, and bench equipment like Function Generators and Oscilloscopes.
- Conducted %GRR analysis on data from bench equipment and automated testers utilizing JMP software.

Graduate Intern

May 2016 - Aug 2016

X-FAB, Lubbock, Texas

Project: Survey of SiC epitaxial deposition equipment and processes

- Researched and evaluated the economic practicality of in-house SiC wafer growth for X-FAB.
- Liaised with SiC epi deposition and characterization tool vendors for technical discussions.
- Conducted research and made SiC epi characterization tool selections as part of a multidisciplinary team.
- Assessed tools against X-FAB specifications, recommending the optimal choice.

Trainee

Apr 2016 - Apr 2016

National Instruments, Austin, Texas

- Performed functional testing of the mixed-signal IC ADS7229 using T-bench.
- Applied dynamic pin map methodologies for testing mixed-signal ICs with the TestStand Semiconductor Module integrated into LabVIEW and NI TestStand.

Teaching Assistant

Sept 2015 - Jan 2016

Texas Tech University, Lubbock, Texas

- Graded coursework, including assignments, exams, and projects, for the Modern Digital Systems course in the ECE department.

Graduate Engineer Trainee

Jun 2014 - Jun 2015

Reliance Communications, India

- Administered databases with basic SQL queries and handled file transfers using UNIX commands.

Projects

Validate the datasheet of Edge Triggered D Flip Flop SN74HC574

Oct 2016 - Dec 2016

- Tested and validated electrical and timing characteristics of D Flip Flops with equipment such as Keithley 2410, Infiniium Oscilloscope, NI STS, and PXI-e 1075.
- Automated LabVIEW code to extract electrical parameters and analyzed results with JMP software to determine process capability metrics such as Cp, Cpk, and %GRR.

Mixed Signal Characterization and Testing of ADC 7800JU

Apr 2016 - May 2016

- Performed comprehensive parametric and functional testing of ADC 7800 with NI STS T4, PXIe 1075, and various bench tools.
- Designed and implemented LabVIEW code and procedures to test and ensure DUT compliance with datasheet parameters.

Parametric Testing of Op Amp LM741

Feb 2016 - Mar 2016

- Performed a range of parametric tests on Op Amps with bench equipment, utilizing both manual and automated methods.
- Conducted test result validation and analyzed data in JMP software to determine metrics such as Cp and Cpk.

Technical Skills

- **Programming Languages:** VBA, Python, C++, SQL, and UNIX.
- **Lab Equipments:** Teradyne UltraFlex and J750, Oscilloscope, NI Semiconductor Test System, NI Virtual Bench, Function Generators, Keithley and Agilent power supplies.
- **Software Tools:** NI LabVIEW, NI TestStand, Teradyne IG-XL, STDF viewer, JIRA, TSSI Wavemakerplus, TIBCO Spotfire, Datapower, Optimal Plus, JMP, MS Office, GitLab, Tortoise SVN, Cadence Virtuoso, PCB Wizard, Autodesk Inventor 2015.